

Insight Report

Assignment 1
EDA Presentation

Data Cleaning

During task 2, I noticed a series of artifacts, errors, and outliers that needed addressing. These were the most important:

- Negative values when they don't make sense
- Inconsistent data input conventions (free vs free trial, TIER1 vs tier1, etc)
- Logged stream duration vs computed duration (logged duration didn't match start and end timestamps)

These major issues were addressed in Task 2. Negative values were turned positive, naming conventions for tiers were fixed, and logged stream duration was replaced with computed duration.

#Streams data

```
#Turn all negative values positive
streams['peak_viewers'] = streams['peak_viewers'].abs()

#Let's see if the new minimum is positive
streams['peak_viewers'].describe().round(2)
```

Turning negative values positive for peak viewers

```
#Compute number of minutes from the start time and end time
computedMins = (sessions['ended_at'] - sessions['started_at']).dt.total_seconds() / 60

#Check if the computed minutes match the logged duration_mins
matches = ((computedMins[checkReversed].abs() - sessions.loc[checkReversed, 'duration_mins']).abs() < 2).all()
print(matches)
```

Replacing logged duration with computed duration

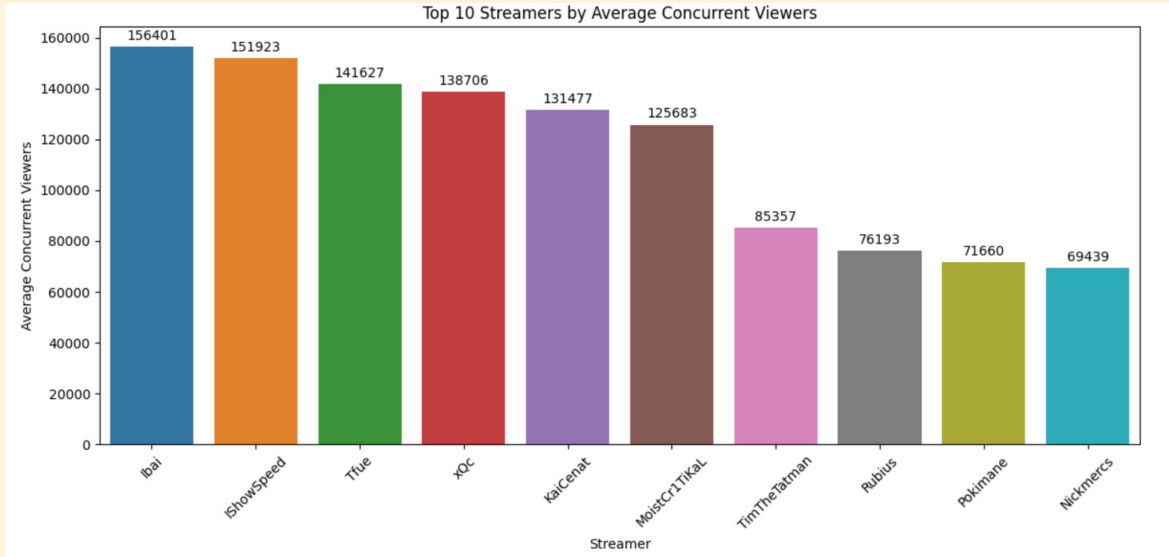
```
#Map messed up values to good values
tierMap = {
    'free trial' : 'free',
    'none' : 'free',
    'TIER1' : 'tier1',
    'tier4' : 'tier3',
    'premium' : 'tier3'
}

#Replace values using the map
viewers['subscription_tier'] = viewers['subscription_tier'].replace(tierMap)

#Check if this worked
print(viewers['subscription_tier'].unique())
```

Fixing tier naming convention

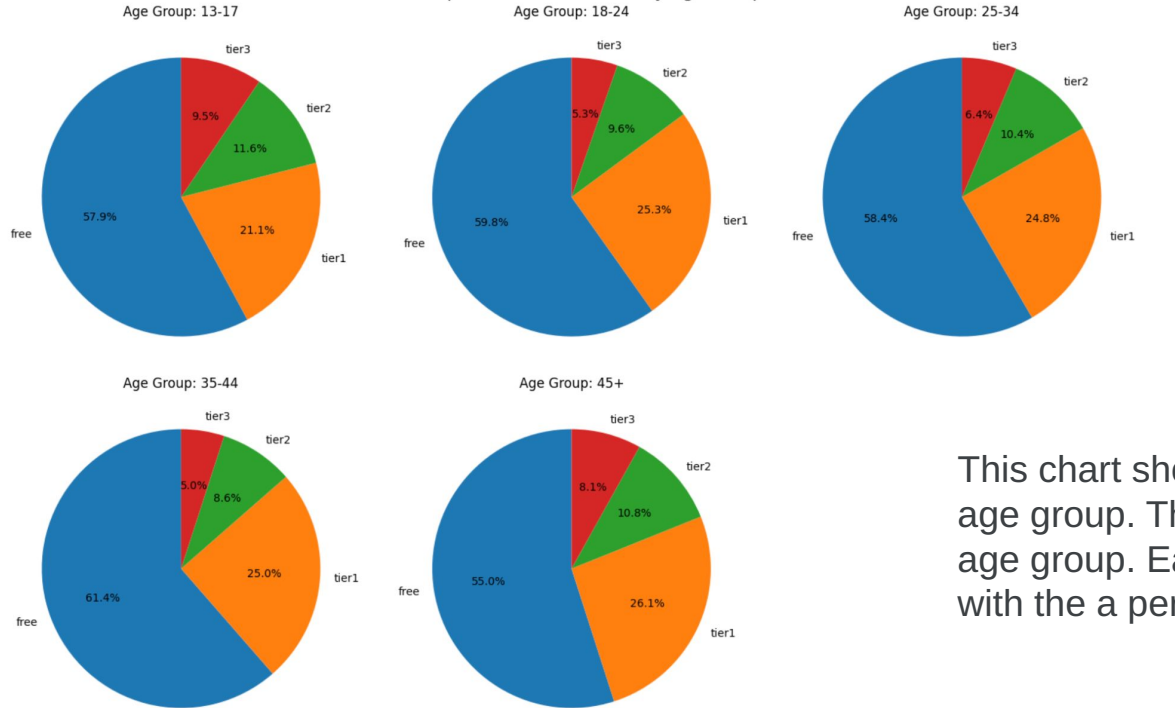
Top Streamers



As a non-Twitch user, I was very interested to see the top ten streamers. This bar graph shows the ten streamers with the greatest number of average concurrent viewers. Ibai has the greatest number of average concurrent viewers at 156,401.

Subscription Tiers and Ages

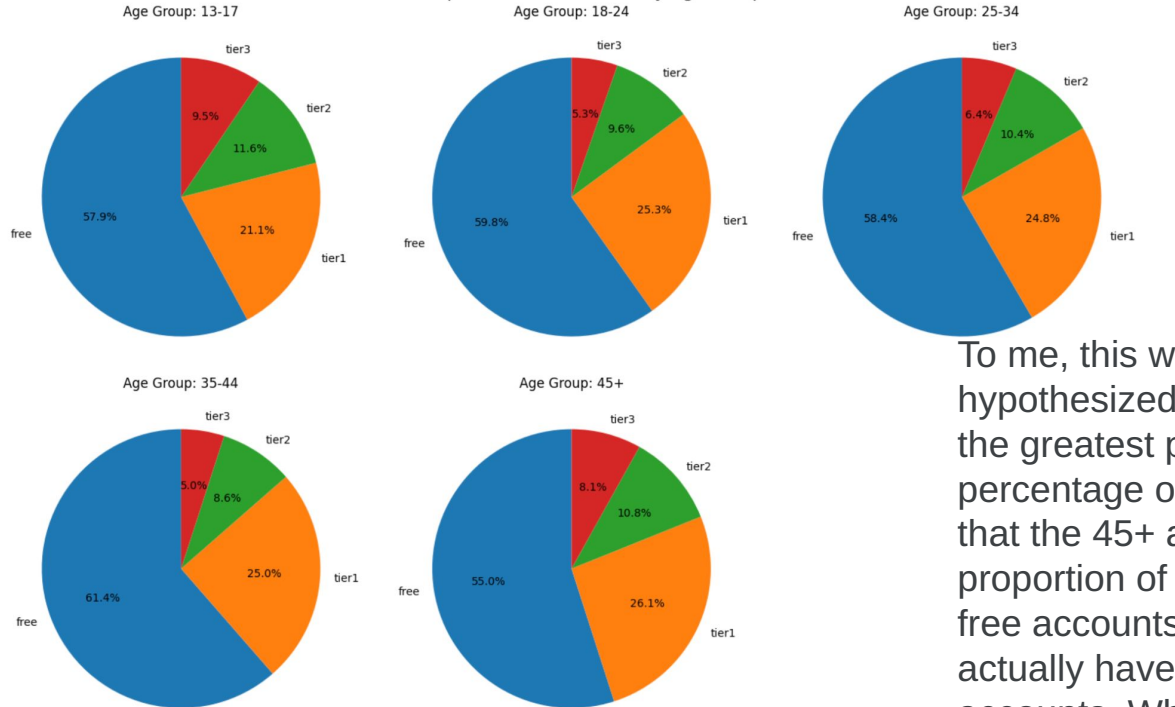
Subscription Tier Distribution by Age Group



This chart shows the subscription tier distribution by age group. This is a series of pie charts, one for each age group. Each section of the pie chart is labeled with the a percentage and tier.

Subscription Tiers and Ages

Subscription Tier Distribution by Age Group



To me, this was the most interesting insight. I had hypothesized that the 13-17 age group would have the greatest percentage of free accounts and lowest percentage of tier 3 accounts. Additionally, I predicted that the 45+ age group would have the highest proportion of tier 3 accounts and lowest percentage of free accounts. Here, we can see that 13-17 year olds actually have the highest proportion of tier 3 accounts. Where are they getting this money?